

Fig 3o. What to run next: a cost-tiered plan that excludes the confound before naming the activity

The drug shows that SOME activity it perturbs is required at 39C; the deck shows the molecular label is not identifiable.
Climb the ladder (bottom -> top): Tier 0 exclude the heat x drug-cytotoxicity confound -> Tier 1 'is it HIF at all?'
-> Tier 2 isoform genetics. Tier 2 is GATED -- do NOT lead with floxed-Hif1a vs floxed-Epas1. (belzutifan = a HIF2a-selective cross-check the drug cannot exclude, not a crowning of HIF2a.)

CONCEPTUAL – not data

Tier 2 – isoform genetics (gated) [cost: expensive (gated)]

Floxed-Hif1a iTreg @ 39C

Tests: does deleting HIF1a (specifically) abolish the 39C effect?

Floxed-Epas1 (HIF2a) iTreg @ 39C

Tests: does deleting HIF2a instead abolish it -- which isoform, if either?

Run ONLY after Tiers 0–1 -- do NOT lead with floxed-Hif1a vs floxed-Epas1 genetics.

Tier 1 – is it HIF at all? [cost: medium]

Chaperone / proteostasis inhibitor @ 39C

Tests: is the required activity HSP90/HSP70 chaperone dependence (the heat-shock module that dominates fig3l)?

Glycolysis inhibitor @ 39C

Tests: whether the shared glycolytic members (one part of the program, NOT the dominant module) carry the dependency

HSF1 readout / knockdown @ 39C

Tests: is the heat-shock axis (HSF1; fig3n co-elevated) the driver rather than HIF?

Belzutifan (HIF2a-selective) cross-check

Tests: an alternative the drug cannot exclude -- is a HIF2a-selective block sufficient?

Run only AFTER Tier 0 excludes the heat x drug-cytotoxicity confound.

Tier 0 – exclude the confound [cost: cheap]

Inactive structural analog / vehicle @ 39C

Excludes: the effect is heat x drug-cytotoxicity, not a specific perturbed activity

Dose-response of the drug @ 39C

Tests: is the 39C effect on-target and dose-dependent, or a cytotoxic cliff?

Tconv comparator @ 39C (HIF-agnostic)

Excludes: the drug also kills/impairs Tconv at 39C (generic heat x drug toxicity)

Start here: licensed immediately (the cheap exclusion the deck demands first).